INCH-POUND

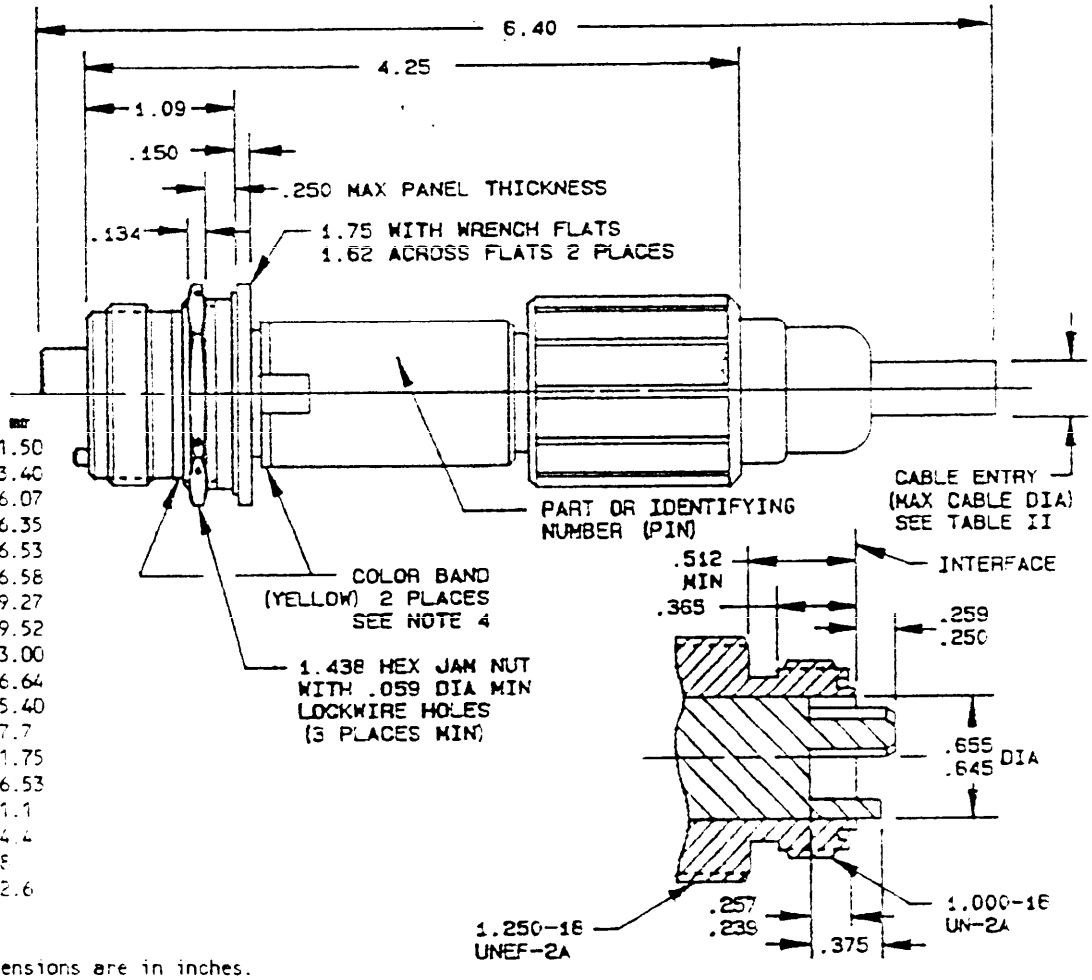
MIL-C-83526/3B
14 Oct 1993
SUPERSEDING
MIL-C-83526/3A
30 June 1989

MILITARY SPECIFICATION SHEET

CONNECTORS, RECEPTACLE, FIBER OPTIC, CIRCULAR, HERMAPHRODITIC, JAM-NUT MOUNTING, 2 AND 4 POSITIONS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83526.



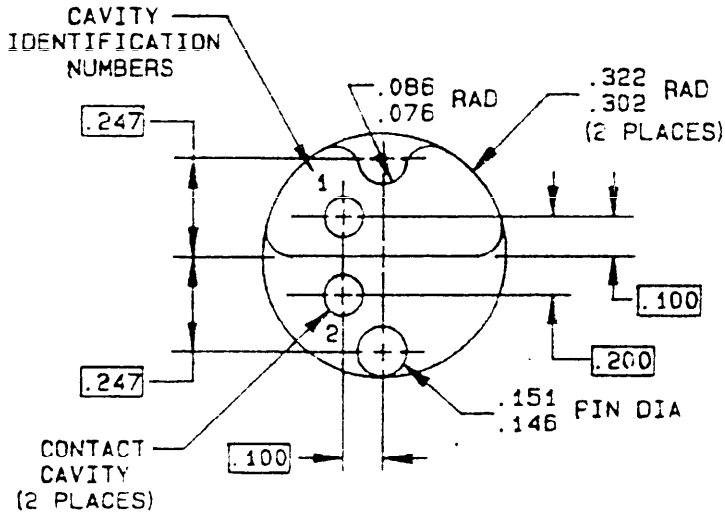
Inches	mm
.059	1.50
.134	3.40
.239	6.07
.250	6.35
.257	6.53
.259	6.58
.365	9.27
.375	9.52
.512	13.00
.655	16.64
1.000	25.40
1.09	27.7
1.250	31.75
1.438	36.53
1.62	41.1
1.75	44.4
4.25	108
6.40	162.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± 0.010 (0.25 mm) for three place decimals and ± 0.03 (0.8 mm) for two place decimals. Angular tolerances are ± 0° 30'.
4. The phrase "Fiber-Optic" must be included within the yellow band.
5. Unless otherwise specified, dimensions are symmetrical about centerlines.

FIGURE 1. Jam-nut connector.

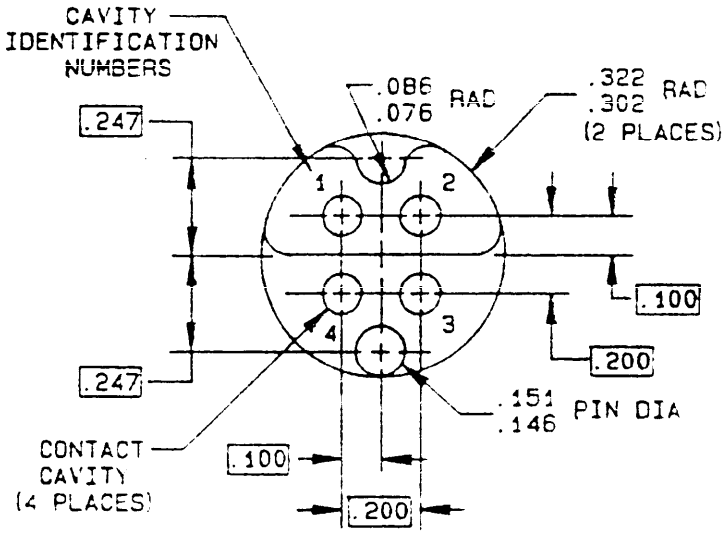
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

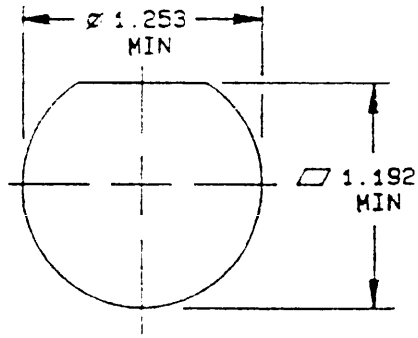
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:37Z
Check the source to verify that this is the current version before use.



Inches	mm
.076	1.93
.086	2.18
.100	2.54
.146	3.71
.151	3.84
.200	5.08
.247	6.27
.302	7.67
.322	8.18
1.192	30.28
1.253	31.83

2 Terminal connector type 3-01





4 Terminal connector type 3-02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Terminal configuration face view.

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figures 1 and 2.

Fiber optic cable:

Cable diameter: In accordance with MIL-C-85045.

Cable configuration: In accordance with MIL-C-85045.

Fiber diameter: In accordance with MIL-F-49291.

Fiber numerical aperture: In accordance with MIL-F-49291.

Fiber type: Multimode.

Fiber class: Graded index.

Fiber composition: Glass/glass.

Materials: The material requirements are listed in table I.

Connector termination and assembly tools: As specified in MIL-K-83533.

Cleaning procedures: The cleaning procedures for the connector shall be included with the assembly instructions of the connector.

Termini: The required number of termini shall be supplied with each connector unless otherwise specified. Style P(pin) or S(socket) see MIL-T-29504/1 and MIL-T-29504/02. For dummy terminus, see MIL-T-29504/3.

Weight: 8.0 ounces maximum.

Coupling forces: 11.0 pounds.

Coupling torques: 5 inch-pounds minimum.

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TABLE I. Material description.

Parts description	Material	Finish
Connector shell Insert (2 termini) Insert (4 termini) Compression nut	Aluminum alloy 2011-T3 QQ-A-225/3 "	Anodized MIL-A-8625, type II class 2, Color: gray
Strain relief Compression sleeve Inner sleeve Strain relief sleeve	" " " "	Chemical film MIL-C-5541, class 1A "
Jam-nut	MS 27512-14E	None
Securing screws Dowel pins	CRES MS 16995-9	Passivated QQ-P-35
Strain relief boot O-rings	Flourosilicone rubber MIL-R-25988, type I, class 1	None None
Shrink tubing	Melt seal polyolefin MIL-I-23053/4	None

PIN:

M83526/03 X X X X

Basic specification sheet

Number of terminus positions required

Quantity	Designator
2	A
4	B

Cable size 1/

Cable OD in inches	Designator
0.178 - 0.216	1
0.217 - 0.256	2

Terminus supplied for actual fiber size 2/

Fiber OD in microns	Designator
123.0 - 124.4	A
124.5 - 127.0	B
138.0 - 139.6	C
139.7 - 142.0	D
No termini supplied	E

Keying position for terminus insert arrangement

Position	Designator
No alternate keying position required	3

1/ Select cable size and type in accordance with MIL-C-85045.
2/ Select fiber sizes and tolerances in accordance with MIL-F-49291.

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Quality conformance and periodic inspection: The following cables shall be used for quality conformance and periodic inspection.

Number channels	Cables
2	PIN M85045/2B2A, 50 micron core, 125 clad, or equal.
4	PIN M85045/2C4A, 100 micron core, 140 clad, or equal.

For quality conformance and periodic inspection, dummy termini shall be used in all unused cavities.

Patent notice: The Government does not have a royalty-free license under the following listed patents for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Patent number	Patent expiration date
US 3,994,564	11/30/1993
US 4,054,366	10/18/1994
US 4,060,308	11/29/1994

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Navy - AS, SH
Air Force - 17, 19
DLA - ES

User activities:
Navy - MC, OS, YD
Air Force - 13

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6060-0100-03)

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